2024 AMC 10A Problem 9 - Solution

Review the full statement and step-by-step solution for 2024 AMC 10A Problem 9. Great practice for AMC 10, AMC 12, AIME, and other math contests

2024 AMC 10A Problem 9

Problem:
In how many ways can 6 juniors and 6 seniors form 3 disjoint teams of 4 people so that each team has 2 juniors and 2 seniors?

Answer Choices:

A. 720

B. 1350

C. 2700

D. 3280

E. 8100

Solution:

Select the first junior and call this person A. There are 5 other juniors and $\binom{6}{2} = 15$ pairs of seniors who could team up with A. Select the next junior not yet on a team, say B. There are 3 other juniors and $\binom{4}{2} = 6$ pairs of seniors who could team up with B. The third team consists of the people not yet chosen. Thus there are $5 \cdot 15 \cdot 3 \cdot 6 = \textbf{(B)}\boxed{1350}$ ways to form the teams.

$$\textbf{OR}$$

There are $\binom{6}{2,2,2} = \frac{6!}{2!2!2!} = 90$ ways to assign the seniors to teams 1,2 , and 3 , and, similarly, 90 ways to assign juniors to those teams. But there are $3! = 6$ ways for the three teams to be ordered, so the number of ways to form the teams is $\frac{90 \cdot 90}{6} = \text{(B)} \boxed{1350}$.